Robustness to Gaussian weight perturbation: muon vs adamw

$\gamma = 0.003$

	30M	60M	100M	300M	600M
128		-0.0044			
64	-0.0099				
32	-0.0216	-0.0129	+0.0035		
16	-0.0221	-0.0178	-0.0012		
8	-0.0300	-0.0289	-0.0176	-0.0116	
4	-0.0488	-0.0301	-0.0270	-0.0145	
2	-0.1088	-0.0828	-0.0448	-0.0260	-0.0171
1	-0.3474	-0.1681	-0.1232	-0.0340	-0.0167
0.5	-0.4580	-0.3976	-0.2449	-0.0705	-0.0316
0.25	-0.6942	-0.7753	-0.6191	-0.1527	-0.0729

$\gamma = 0.005$

	30M	60M	100M	300M	600M
128		-0.0036			
64	-0.0082				
32	-0.0206	-0.0124	+0.0042		
16	-0.0214	-0.0174	-0.0006		
8	-0.0294	-0.0286	-0.0172	-0.0115	
4	-0.0484	-0.0297	-0.0268	-0.0145	
2	-0.1085	-0.0827	-0.0446	-0.0259	-0.0170
1	-0.3472	-0.1681	-0.1231	-0.0339	-0.0167
0.5	-0.4579	-0.3975	-0.2448	-0.0705	-0.0315
0.25	-0.6941	-0.7753	-0.6191	-0.1527	-0.0729

$\gamma = 0.007$

	30M	60M	100M	300M	600M
128		-0.0023			
64	-0.0060				
32	-0.0193	-0.0117	+0.0053		
16	-0.0206	-0.0169	+0.0003		
8	-0.0287	-0.0281	-0.0166	-0.0115	
4	-0.0482	-0.0292	-0.0266	-0.0145	
2	-0.1082	-0.0824	-0.0444	-0.0258	-0.0169
1	-0.3469	-0.1679	-0.1229	-0.0339	-0.0166
0.5	-0.4577	-0.3974	-0.2448	-0.0705	-0.0315
0.25	-0.6942	-0.7751	-0.6190	-0.1526	-0.0729

$\gamma = 0.01$

	30M	60M	100M	300M	600M
128		+0.0003			
64	-0.0018				
32	-0.0164	-0.0100	+0.0075		
16	-0.0188	-0.0156	+0.0021		
8	-0.0272	-0.0271	-0.0154	-0.0114	
4	-0.0475	-0.0281	-0.0261	-0.0143	
2	-0.1075	-0.0820	-0.0440	-0.0256	-0.0166
1	-0.3464	-0.1676	-0.1226	-0.0338	-0.0164
0.5	-0.4574	-0.3972	-0.2446	-0.0704	-0.0314
0.25	-0.6939	-0.7749	-0.6189	-0.1526	-0.0728

$\gamma = 0.02$

	30M	60M	100M	300M	600M
128		+0.0162			
64	+0.0271				
32	+0.0011	-0.0006	+0.0213		
16	-0.0074	-0.0085	+0.0132		
8	-0.0183	-0.0213	-0.0082	-0.0106	
4	-0.0430	-0.0218	-0.0234	-0.0133	
2	-0.1031	-0.0794	-0.0414	-0.0244	-0.0149
1	-0.3432	-0.1658	-0.1208	-0.0331	-0.0156
0.5	-0.4551	-0.3958	-0.2437	-0.0698	-0.0308
0.25	-0.6929	-0.7738	-0.6181	-0.1523	-0.0725

$\gamma = 0.03$

	30M	60M	100M	300M	600M
128		+0.0449			
64	+0.0740				
32	+0.0301	+0.0160	+0.0454		
16	+0.0112	+0.0037	+0.0335		
8	-0.0032	-0.0114	+0.0039	-0.0098	
4	-0.0355	-0.0111	-0.0186	-0.0117	
2	-0.0955	-0.0750	-0.0369	-0.0223	-0.0122
1	-0.3378	-0.1629	-0.1177	-0.0321	-0.0141
0.5	-0.4512	-0.3934	-0.2422	-0.0688	-0.0297
0.25	-0.6911	-0.7723	-0.6167	-0.1518	-0.0719

$\gamma = 0.04$

	30M	60M	100M	300M	600M
128		+0.0885			
64	+0.1440				
32	+0.0711	+0.0395	+0.0812		
16	+0.0391	+0.0218	+0.0628		
8	+0.0181	+0.0034	+0.0228	-0.0084	
4	-0.0245	+0.0042	-0.0117	-0.0084	
2	-0.0844	-0.0688	-0.0304	-0.0194	-0.0078
1	-0.3306	-0.1585	-0.1131	-0.0307	-0.0114
0.5	-0.4458	-0.3900	-0.2401	-0.0674	-0.0280
0.25	-0.6883	-0.7701	-0.6147	-0.1511	-0.0709

$\gamma = 0.05$

	30M	60M	100M	300M	600M
128		+0.1506			
64	+0.2363				
32	+0.1282	+0.0719	+0.1303		
16	+0.0748	+0.0458	+0.1033		
8	+0.0448	+0.0228	+0.0470	-0.0062	
4	-0.0095	+0.0245	-0.0027	-0.0047	
2	-0.0701	-0.0607	-0.0216	-0.0153	-0.0022
1	-0.3208	-0.1528	-0.1071	-0.0288	-0.0084
0.5	-0.4387	-0.3856	-0.2372	-0.0653	-0.0261
0.25	-0.6846	-0.7674	-0.6121	-0.1503	-0.0697

$\gamma = 0.07$

	30M	60M	100M	300M	600M
128		+0.3449			
64	+0.4814				
32	+0.2769	+0.1656	+0.2823		
16	+0.1706	+0.1168	+0.2266		
8	+0.1185	+0.0805	+0.1196	-0.0003	
4	+0.0323	+0.0825	+0.0241	+0.0099	
2	-0.0323	-0.0386	+0.0034	-0.0038	+0.0159
1	-0.2950	-0.1360	-0.0895	-0.0237	+0.0015
0.5	-0.4200	-0.3731	-0.2292	-0.0597	-0.0198
0.25	-0.6751	-0.7598	-0.6049	-0.1477	-0.0660

$\gamma = 0.1$

	30M	60M	100M	300M	600M
128		+0.8150			
64	+0.9478				
32	+0.6060	+0.3934	+0.6907		
16	+0.3724	+0.2892	+0.5533		
8	+0.2764	+0.2209	+0.3047	+0.0177	
4	+0.1289	+0.2255	+0.0930	+0.0661	
2	+0.0555	+0.0136	+0.0651	+0.0259	+0.0658
1	-0.2395	-0.0928	-0.0457	-0.0111	+0.0303
0.5	-0.3796	-0.3429	-0.2102	-0.0459	-0.0040
0.25	-0.6540	-0.7432	-0.5885	-0.1418	-0.0582

$\gamma = 0.13$

	30M	60M	100M	300M	600M
128		+1.3489			
64	+1.4102				
32	+1.0221	+0.7410	+1.3737		
16	+0.5929	+0.5547	+1.0798		
8	+0.4828	+0.4294	+0.6280	+0.0401	
4	+0.2699	+0.4507	+0.2065	+0.2276	
2	+0.1839	+0.0916	+0.1701	+0.0781	+0.1607
1	-0.1632	-0.0190	+0.0295	+0.0106	+0.0873
0.5	-0.3243	-0.2941	-0.1802	-0.0220	+0.0235
0.25	-0.6232	-0.7189	-0.5632	-0.1318	-0.0441

cell = perturbed held-out DCLM loss, tuned muon base minus tuned adamw base | negative (blue) = muon degrades less | white = no data

colour scale clipped at ±0.773 (97th pct); 12 cell(s) beyond it keep their printed value